

Size, Value and a Test of the CAPM

Evidence from the Fama–French 25 Portfolios

Asset Allocation & Investment Strategies: ELE Group 3

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Part I: Evaluating the Single-Factor CAPM

In this part, we evaluate the performance of the single-factor Capital Asset Pricing Model (CAPM) using the 25 Fama–French size–book-to-market portfolios.

Specifically, we:

- Construct excess returns for each portfolio and the market factor
- Estimate time-series CAPM regressions for each of the 25 portfolios
- Report mean excess returns, CAPM betas, alphas, t-statistics, and adjusted R² values
- Examine whether the CAPM can explain the cross-section of portfolio returns

Step 1: Loading the 25 Fama–French Portfolio Returns

The file `25_Portfolios_5x5.csv` contains multiple data sections. Here, we extract the monthly **value-weighted** returns for the 25 portfolios sorted by firm size and book-to-market ratio.

The following steps are performed:

- Skip header rows to reach the monthly return data
- Retain only the monthly value-weighted portfolio block
- Convert all return columns to numeric values
- Handle missing-value codes
- Construct a proper monthly date index

Step 2: Loading Fama–French Factor Data

We load the Fama–French factor dataset containing monthly returns for:

- Mkt–RF: the market excess return
- RF: the one-month risk-free rate

Only the monthly factor block is retained, excluding annual data. All series are converted to numeric format and indexed by date.

	Mkt-RF	SMB	HML	RF
date				
1926-07-01	2.96	-2.56	-2.43	0.22
1926-08-01	2.64	-1.17	3.82	0.25
1926-09-01	0.36	-1.40	0.13	0.23
1926-10-01	-3.24	-0.09	0.70	0.32
1926-11-01	2.53	-0.10	-0.51	0.31

Step 3: Sample Period Selection

To replicate the original Fama–French (1993) analysis, we restrict the sample to January 1964 through January 1993, as specified in the assignment.

Both portfolio returns and factor data are trimmed to this period and aligned on common dates to ensure consistent regression samples.

Sample window: 1964-01-01 to 1993-01-01

Number of months (N): 349

Units: returns are in percent per month (as provided by FF data files).

Step 4: Constructing Excess Returns and Estimating the CAPM

For each portfolio, excess returns are constructed as:

$$r_t^i = R_t^i - RF_t$$

where RF_t is the one-month risk-free rate.

This ensures consistency with the CAPM, which is defined in excess-return form.

Using these excess returns, we estimate the CAPM time-series regression for each of the 25 portfolios:

$$r_t^i = \alpha_i + \beta_i(Mkt - RF)_t + \varepsilon_t^i$$

The regression is estimated using ordinary least squares (OLS).

For each portfolio, we report:

- the mean monthly excess return
- the estimated CAPM beta and its t-statistic
- the estimated alpha and its t-statistic
- the adjusted R^2

Portfolio	Mean r_i (monthly)	beta	t(beta)	alpha	t(alpha)	p(alpha)	Adj R^2
SMALL LoBM	0.308968	1.425884	27.798851	-0.263673	-1.126317	0.260810	0.689224
ME1 BM2	0.742515	1.250530	28.182754	0.240296	1.186556	0.236214	0.695076
ME1 BM3	0.783957	1.158853	28.657677	0.318556	1.726040	0.085230	0.702122
ME1 BM4	0.943715	1.077028	26.867342	0.511176	2.793956	0.005496	0.674418
SMALL HiBM	1.084405	1.103020	24.185137	0.641427	3.081514	0.002224	0.626578
ME2 BM1	0.404740	1.428427	37.147215	-0.168923	-0.962518	0.336460	0.798485
ME2 BM2	0.658241	1.234521	37.442397	0.162452	1.079549	0.281093	0.801022
ME2 BM3	0.882838	1.116015	35.597026	0.434641	3.037569	0.002566	0.784406
ME2 BM4	0.956548	1.033107	34.744860	0.541647	3.991288	0.000080	0.776091
ME2 BM5	1.075462	1.123314	30.086892	0.624334	3.663913	0.000287	0.722094
ME3 BM1	0.449140	1.353789	45.604053	-0.094548	-0.697838	0.485746	0.856597
ME3 BM2	0.697523	1.162876	45.617558	0.230506	1.981224	0.048355	0.856670
ME3 BM3	0.704935	1.033577	41.500499	0.289846	2.549935	0.011204	0.831826
ME3 BM4	0.879180	0.975694	39.040635	0.487337	4.272519	0.000025	0.814020
ME3 BM5	1.020204	1.069104	30.741266	0.590847	3.722447	0.000230	0.730655
ME4 BM1	0.465701	1.223638	55.263285	-0.025718	-0.254490	0.799268	0.897678
ME4 BM2	0.403485	1.128797	57.108716	-0.049845	-0.552534	0.580938	0.903559
ME4 BM3	0.617670	1.034855	49.793979	0.202068	2.130326	0.033848	0.876877
ME4 BM4	0.806612	0.973928	39.517688	0.415478	3.693725	0.000257	0.817672
ME4 BM5	0.919441	1.081046	33.602720	0.485288	3.305077	0.001049	0.764250
BIG LoBM	0.326350	1.001999	49.686748	-0.076058	-0.826358	0.409169	0.876410
ME5 BM2	0.346699	0.982546	58.557294	-0.047897	-0.625438	0.532095	0.907838
ME5 BM3	0.348962	0.866932	40.036931	0.000798	0.008073	0.993563	0.821534
ME5 BM4	0.513917	0.834828	36.661585	0.178646	1.718931	0.086519	0.794213
BIG HiBM	0.612476	0.870462	26.605966	0.262894	1.760606	0.079186	0.670104

Step 5: Formatting Results into 5×5 Size–Book-to-Market Tables

The CAPM regressions produce one row of results per portfolio. To match the presentation format in the assignment (and in Fama–French style), we:

- Map each portfolio name (e.g., ME3 BM4, SMALL LoBM, BIG HiBM) to its 5×5 grid position:
 - rows = size buckets (Small, 2, 3, 4, Big)
 - columns = book-to-market buckets (Low b/m, 2, 3, 4, High b/m)
- Pivot the regression output into separate 5×5 tables for mean excess returns, beta, t(beta), alpha, t(alpha) and adjusted R^2
- Rename row/column labels to match the assignment table template.

Step 6: Highlighting Statistically Significant CAPM Alphas

To make CAPM mispricing patterns easier to interpret across the 5×5 grid, we highlight statistically significant alphas using conventional significance levels based on the alpha t-statistics:

- * $\rightarrow p < 0.1$
- ** $\rightarrow p < 0.05$
- *** $\rightarrow p < 0.01$

We then display the final set of 5×5 tables required for Part I

bm size	Low b/m	2	3	4	High b/m
Small	-0.264	0.240	0.319*	0.511***	0.641***
2	-0.169	0.162	0.435***	0.542***	0.624***
3	-0.095	0.231**	0.290**	0.487***	0.591***
4	-0.026	-0.050	0.202**	0.415***	0.485***
Big	-0.076	-0.048	0.001	0.179*	0.263*

Visual Evidence: CAPM Fit Across Portfolios

To complement the regression results, we examine the relationship between average excess returns and CAPM betas across the 25 portfolios. Under the CAPM, portfolios with higher betas should earn higher average returns.

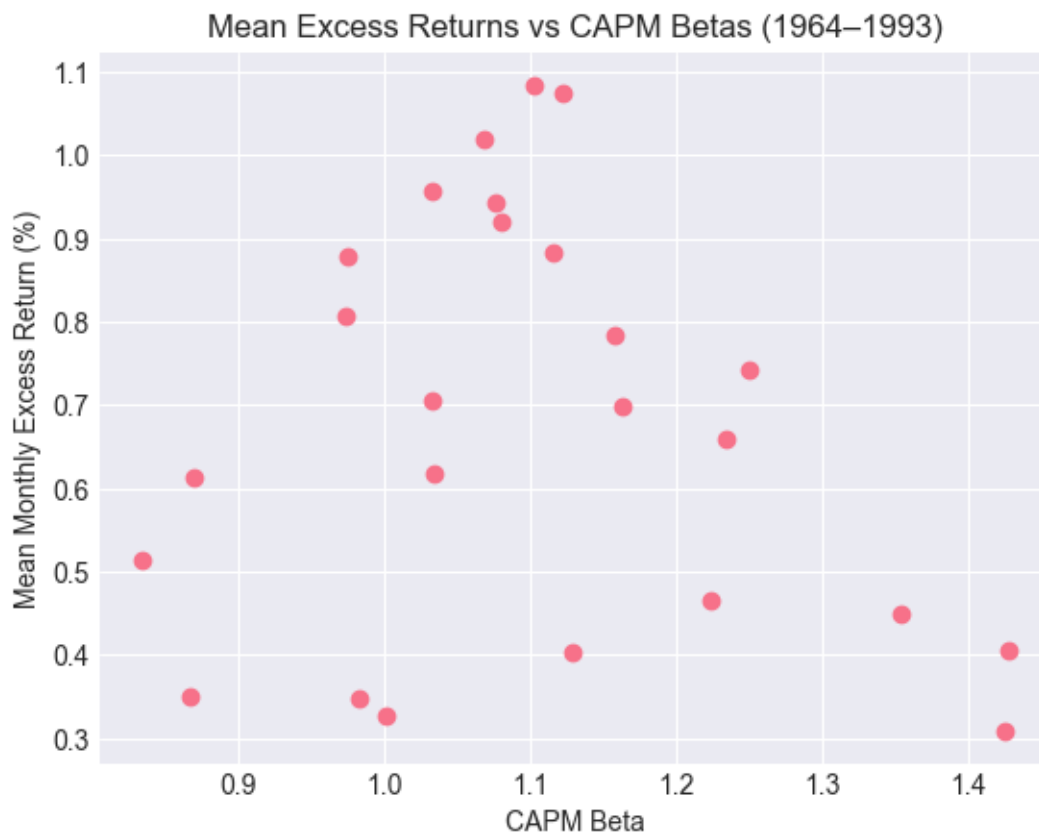


Figure: Mean Excess Returns vs CAPM Betas (1964–1993)

The figure plots average monthly excess returns against CAPM betas for the 25 size–book-to-market portfolios. Under the CAPM, portfolios with higher market exposure (higher betas) should earn higher average excess returns.

Instead, the scatter shows a weak and noisy relationship between betas and mean returns. Portfolios with similar betas often exhibit substantially different average returns, and some high-beta portfolios earn relatively low excess returns. This visual evidence reinforces the regression results and highlights the inability of the single-factor CAPM to explain the cross-section of returns, suggesting the presence of additional systematic risk factors beyond market exposure.

CAPM Results Summary (5×5 Portfolios)

The tables below summarise the CAPM estimation results for the 25 size–book-to-market portfolios, organised in 5×5 format.

Specifically, we report: - mean monthly excess returns - CAPM betas and their t-statistics - CAPM alphas and their t-statistics - adjusted R^2 values

The final table highlights statistically significant alphas using conventional significance levels.

'Mean excess returns (% per month)'

bm size	Low b/m	2	3	4	High b/m
Small	0.309	0.743	0.784	0.944	1.084
2	0.405	0.658	0.883	0.957	1.075
3	0.449	0.698	0.705	0.879	1.020
4	0.466	0.403	0.618	0.807	0.919
Big	0.326	0.347	0.349	0.514	0.612

'CAPM betas'

bm size	Low b/m	2	3	4	High b/m
Small	1.426	1.251	1.159	1.077	1.103
2	1.428	1.235	1.116	1.033	1.123
3	1.354	1.163	1.034	0.976	1.069
4	1.224	1.129	1.035	0.974	1.081
Big	1.002	0.983	0.867	0.835	0.870

't-statistics for betas'

bm size	Low b/m	2	3	4	High b/m
Small	27.80	28.18	28.66	26.87	24.19
2	37.15	37.44	35.60	34.74	30.09
3	45.60	45.62	41.50	39.04	30.74
4	55.26	57.11	49.79	39.52	33.60
Big	49.69	58.56	40.04	36.66	26.61

'CAPM alphas (% per month)'

bm size	Low b/m	2	3	4	High b/m
Small	-0.264	0.240	0.319	0.511	0.641
2	-0.169	0.162	0.435	0.542	0.624
3	-0.095	0.231	0.290	0.487	0.591
4	-0.026	-0.050	0.202	0.415	0.485
Big	-0.076	-0.048	0.001	0.179	0.263

't-statistics for alphas'

bm size	Low b/m	2	3	4	High b/m
Small	-1.13	1.19	1.73	2.79	3.08
2	-0.96	1.08	3.04	3.99	3.66
3	-0.70	1.98	2.55	4.27	3.72
4	-0.25	-0.55	2.13	3.69	3.31
Big	-0.83	-0.63	0.01	1.72	1.76

'Adjusted R²'

bm size	Low b/m	2	3	4	High b/m
Small	0.689	0.695	0.702	0.674	0.627
2	0.798	0.801	0.784	0.776	0.722
3	0.857	0.857	0.832	0.814	0.731
4	0.898	0.904	0.877	0.818	0.764
Big	0.876	0.908	0.822	0.794	0.670

'Alphas with significance stars'

bm size	Low b/m	2	3	4	High b/m
Small	-0.264	0.240	0.319*	0.511***	0.641***
2	-0.169	0.162	0.435***	0.542***	0.624***
3	-0.095	0.231**	0.290**	0.487***	0.591***
4	-0.026	-0.050	0.202**	0.415***	0.485***
Big	-0.076	-0.048	0.001	0.179*	0.263*

Part I Summary: Performance of the Single-Factor CAPM

Methodology Recap

We began by constructing monthly excess returns for each portfolio by subtracting the one-month risk-free rate. Using these excess returns, we estimated CAPM time-series regressions over the period January 1964 to January 1993, replicating the original sample studied by Fama and French (1993).

For each portfolio, we estimated:

- CAPM implied beta
- CAPM alpha, capturing abnormal returns unexplained by market risk
- corresponding t-statistics
- R^2 and the adjusted R^2

To facilitate interpretation, all results were organized into 5×5 tables based on portfolio size and book-to-market characteristics, matching the standard Fama–French presentation format.

Empirical Findings

The estimated CAPM betas are economically sensible and highly statistically significant across all portfolios, indicating strong exposure to the market factor. Adjusted (R^2) values are relatively high, confirming that market returns explain a significant fraction of time-series variation in portfolio returns.

However, despite this strong time-series fit, the CAPM performs poorly in explaining the cross-section of average returns. Portfolios with similar betas often exhibit markedly different mean excess returns, as illustrated both in the regression tables and in the scatter plot of mean excess returns versus CAPM betas.

Most notably, high book-to-market portfolios consistently display positive and often statistically significant CAPM alphas, while low book-to-market portfolios tend to exhibit negative or insignificant alphas. These patterns persist across size categories and are especially pronounced among smaller firms.

Interpretation

The presence of systematic and economically meaningful alphas indicates that market risk alone is insufficient to explain expected returns. While the CAPM captures common market movements, it fails to account for the strong return differentials associated with size and book-to-market characteristics.

These findings replicate the central empirical puzzle highlighted by Fama and French (1993) and motivate the inclusion of additional risk factors beyond the market portfolio. In the follow parts of the assignment, we explore whether multifactor models can better explain these cross-sectional return patterns.

Part II: Interpreting Patterns in Average Excess Returns (5×5 Portfolios)

In this part, we discuss the patterns in average excess returns across the 25 Fama–French size–book-to-market portfolios. The goal is to translate the 5×5 return table into an economic description of how expected returns vary with firm size and btm.

The portfolios are arranged as a 5×5 grid:

- **Rows (North/South)** represent size buckets (Small at the top, Big at the bottom).
- **Columns (West/East)** represent book-to-market buckets (Low b/m on the left, High b/m on the right).

We will: - Describe how mean excess returns change as we move **East/West** (value dimension) and **North/South** (size dimension). - Quantify these directional patterns using simple spreads computed from the mean excess return table:

- **Value spread (within each size row):** High b/m minus Low b/m
- **Size spread (within each b/m column):** Small minus Big
- Use these spreads to summarize whether returns increase more strongly with book-to-market, with size, or through an interaction of both.

Step 1: Quantifying Directional Return Patterns (Value and Size Spreads)

To complement the qualitative discussion of “North/South/East/West” return patterns in the 5×5 grid, we compute simple return spreads from the mean excess return table.

Value Spread

Value spread (East – West) within each size row

$$\text{ValueSpread}_{\text{size}} = \bar{r}(\text{High b/m}) - \bar{r}(\text{Low b/m})$$

This measures how much average excess returns increase as we move from **Low b/m** to **High b/m**, holding size fixed.

	High b/m – Low b/m
size	
Small	0.775
2	0.671
3	0.571
4	0.454
Big	0.286

Size Spread

Size spread (North – South) within each book-to-market column

$$\text{SizeSpread}_{b/m} = \bar{r}(\text{Small}) - \bar{r}(\text{Big})$$

This measures how much average excess returns increase as we move from **Big** to **Small**, holding book-to-market fixed.

These spreads provide a compact numerical summary of whether the dominant return variation is driven primarily by the value dimension (East/West), the size dimension (North/South), or an interaction of both.

bm	Low b/m	2	3	4	High b/m
Small – Big	-0.017	0.396	0.435	0.43	0.472

Discussion of Excess-Return Patterns in the 5×5 Grid (North/South/East/West)

Part II asks us to describe, in words, how **average excess returns vary across the 5×5 portfolios** and to state **in which directions** (North/South/East/West) returns increase.

- **Rows (North/South):** Size buckets, with **Small at the top** and **Big at the bottom**.
Moving **North** means moving toward **smaller** firms; moving **South** means moving toward **bigger** firms.
- **Columns (West/East):** Book-to-market buckets, with **Low b/m at the left** and **High b/m at the right**.
Moving **East** means moving toward **higher book-to-market** (“value”); moving **West** means moving toward **lower book-to-market** (“growth”).

East–West pattern (book-to-market / value dimension)

The dominant pattern is a strong **Eastward increase** in mean excess returns: within *every* size row, average excess returns rise as we move from **Low b/m** → **High b/m**.

This is visible directly in the mean excess return table (e.g., for Small: **0.309** → **1.084**), and it is confirmed numerically by the **value spreads** (High b/m – Low b/m), which are positive in every size row:

- Small: **0.775**
- 2: **0.671**
- 3: **0.571**
- 4: **0.454**
- Big: **0.286**

So, **returns increase to the East** across the entire grid, implying a clear **value premium**.

A second important feature is that the East–West gradient is **steepest in the North**: the value spread is largest among Small firms and declines monotonically as we move South toward Big firms. This indicates an interaction where the **value effect is strongest for smaller firms**.

North–South pattern (size dimension)

The North–South pattern is **not uniform** across all columns; it depends on book-to-market.

To quantify this, we compare **Small — Big** within each book-to-market column:

- Low b/m: **−0.017**
- 2: **0.396**
- 3: **0.435**
- 4: **0.430**
- High b/m: **0.472**

This shows that: - In the **Low b/m (growth) column**, there is essentially **no size premium** (Small is slightly *lower* than Big). - For **middle-to-high b/m columns**, returns are **higher in the North** (Small > Big), and the size premium becomes economically meaningful.

So, **returns increase to the North primarily in the middle-to-high book-to-market portfolios**, whereas in low b/m portfolios the North–South gradient is weak or absent.

Overall directional conclusion

Putting both dimensions together:

- The clearest and most consistent pattern is that **returns increase to the East** (toward higher book-to-market).
- A secondary pattern is that **returns often increase to the North** (toward smaller firms), but mainly in the **value** (higher b/m) region of the grid.
- The combined evidence implies that the cross-section of excess returns is driven primarily by a strong **value effect**, with a **size effect that is conditional on book-to-market** (i.e. an interaction between the two characteristics).

Part III: The CAPM Puzzle

The Core Puzzle Explained

Under the CAPM, expected returns should be determined entirely by market risk (beta):

$$E[R_i] = R_f + \beta_i \cdot (E[R_m] - R_f)$$

This implies three testable predictions:

1. The relationship between beta and returns should be positive and linear
2. Beta should be the only variable explaining cross-sectional return differences
3. CAPM alphas (abnormal returns) should be zero for all portfolios

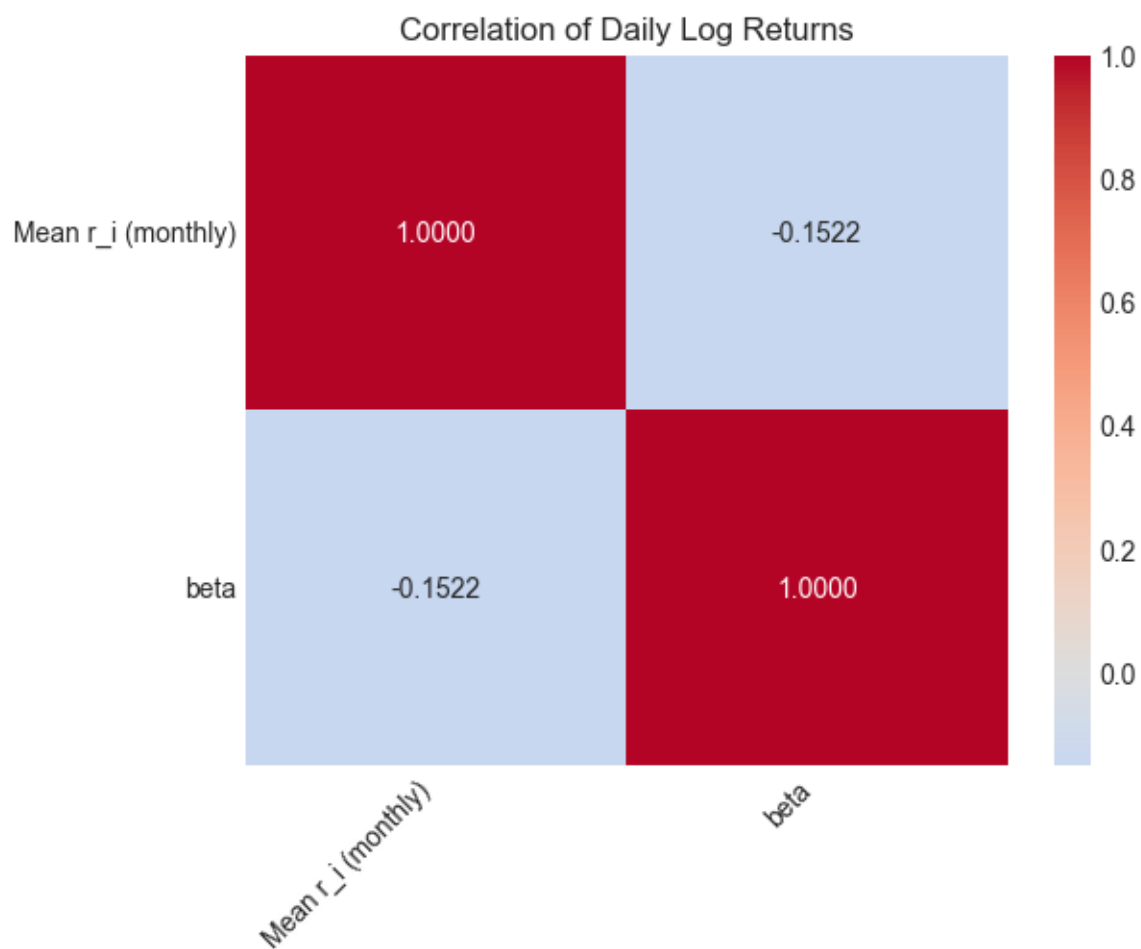
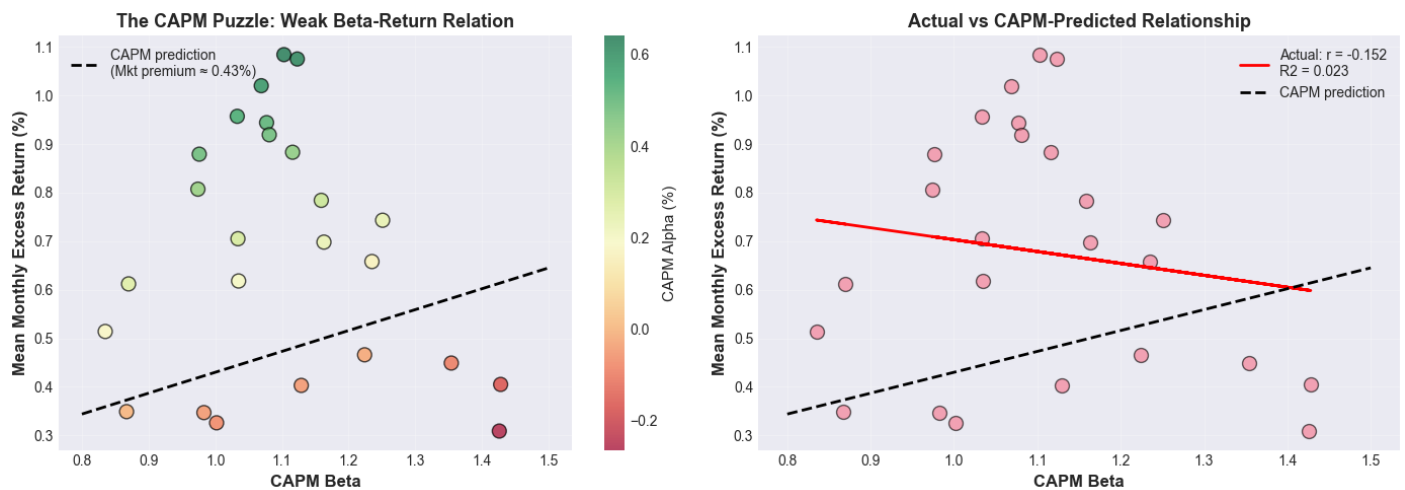


Table 13

	Statistic	Value	CAPM Prediction
0	Correlation (Beta, return)	-0.1524	Strongly positive (>0.8)
1	R2 (cross-section)	0.0232 (2.3%)	High ($\sim 0.9+$)
2	Return range	0.31% to 1.08%	Explained by beta
3	Beta range	0.83 to 1.43	Explains returns
4	Return spread	0.78%	Proportional to beta spread
5	Beta spread	0.59	Proportional to return spread

As we can see from this initial plot, the β only explains 2.3% of the cross-sectional variation of returns. To make matters worse, the empirical β is in fact weakly negatively correlated; this seems like damning evidence against the CAPM's efficacy in Portfolio Analysis.



Clear evidence against the CAPM comes from direct portfolio comparisons:

Table 14

	Portfolio	Beta	Mean Return (%)	Interpretation
0	Small High B/M	1.103000	1.084	
1	ME2 Low B/M	1.428000	0.405	
2	Difference	-0.325000	0.679	30% lower Beta, 164% higher return
3	Big High B/M	0.870000	0.612	
4	ME4 Low B/M	1.224000	0.466	
5	Difference	-0.354000	0.146	29% lower Beta, 31% higher return

The Beta Pattern Contradicts Return Patterns

Value Effect: Returns Rise, Betas Fall

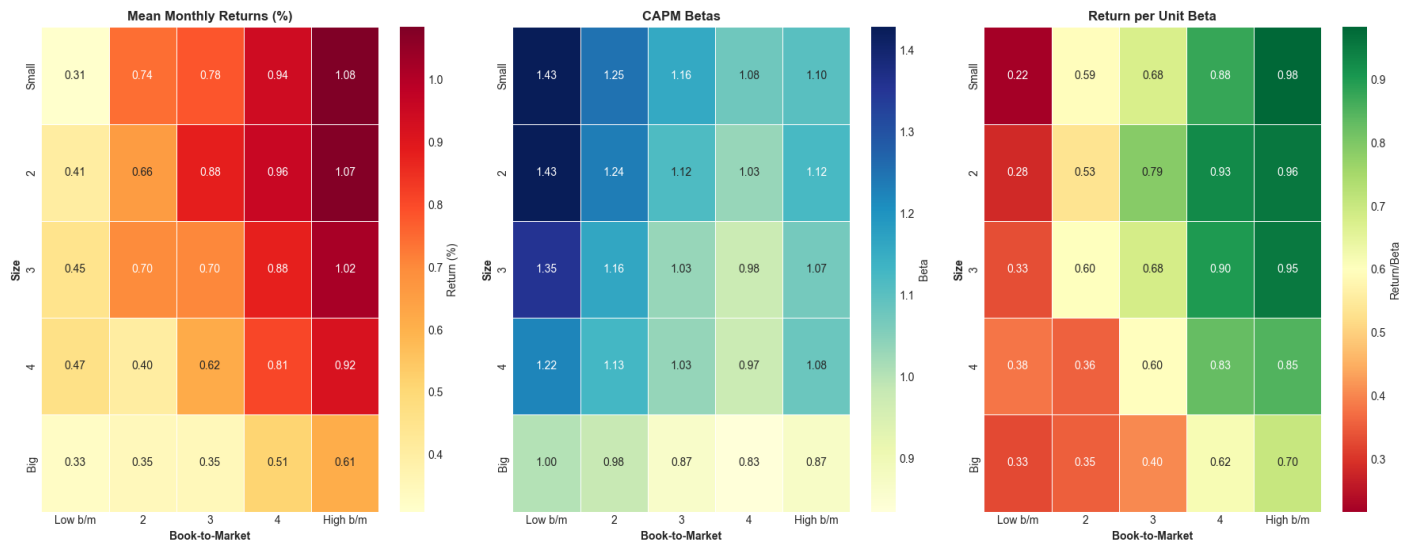
Table 15

	Return Spread (High B/M - Low B/M)	Beta Spread (High B/M - Low B/M)	Ratio (Return/Beta)
Small	+0.775%	-0.323	2.399
2	+0.670%	-0.305	2.197
3	+0.571%	-0.285	2.004
4	+0.453%	-0.143	3.168
Big	+0.286%	-0.132	2.167

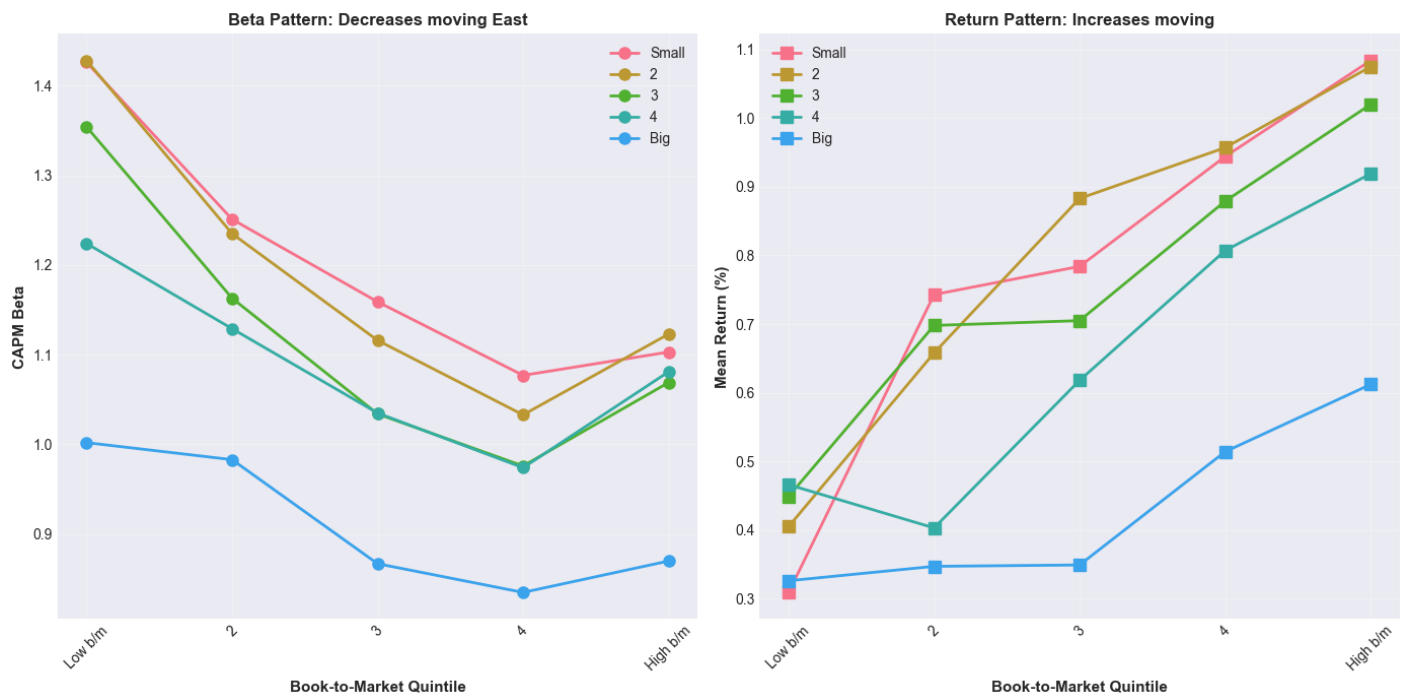
This is critical. Here even as β decreases, returns increase



Return/Beta Mismatch



Pattern Analysis: Beta vs Returns over BTM



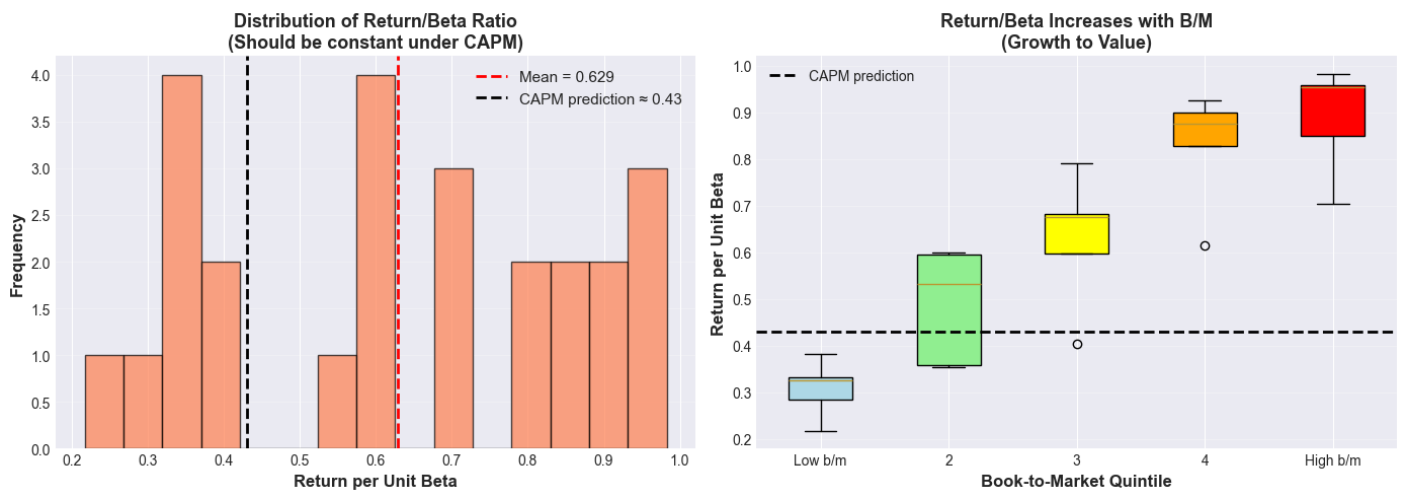
As we move toward value stocks (East), betas decline while returns rise. This contradicts the CAPM's prediction of a positive beta-return relationship.

Quantifying the Puzzle

Return/Beta Ratio Analysis

Table 16

	Metric	Value	CAPM Prediction
0	Mean return/Beta ratio	0.6288	Constant near 0.43
1	Min return/Beta ratio	0.2167	Same as mean
2	Max return/Beta ratio	0.9828	Same as mean
3	Std dev of return/Beta	0.2394	Near zero
4	Coefficient of variation	0.3808	Near zero
5	Range (max - min)	0.7661	Near zero



Time-Series vs Cross-Section

Table 17

	Dimension	Metric	Mean Value	Range	Result
0	Time-Series	Adj R2 (within portfolios)	0.788	0.627 - 0.908	Explains 63-91% of variation
1	Cross-Section	R2 (across portfolios)	0.023	N/A	Explains only 2.3% of variation

Violation Summary

Table 18

	CAPM Prediction	Actual Result
0	Positive Beta-Return correlation	Negative correlation (-0.152)
1	High R ² in cross-section	R ² = 0.023 (2.3%)
2	Alphas = 0	8 out of 25 significantly different to zero
3	Constant Return/Beta ratio	Range: 0.217 to 0.983
4	Returns up when Beta up	Returns up when Beta down (value effect)
5	Beta explains all return variation	B/M and size matter more than Beta

Practical Implications

The CAPM puzzle has important implications for:

- Portfolio Construction: Beta alone is not enough for risk assessment; size and value exposures matter
- Cost of Capital: Using CAPM will systematically misprice securities, especially value vs. growth stocks
- Performance Evaluation: CAPM alphas may reflect compensation for missing risk factors (size, value) rather than manager skill
- Asset Pricing Theory: Market portfolio is likely not mean-variance efficient; multiple risk factors are needed (extension of Rolls Critique)

Conclusion and Takeaways

The spread in average returns across the 25 Fama-French portfolios represents a fundamental rejection of the single-factor CAPM. The evidence is overwhelming:

- Quantitative: Beta explains only 2.3% of cross-sectional return variation
- Directional: The beta-return relationship is negative, not positive
- Systematic: High B/M portfolios consistently have lower β 's but higher returns

This puzzle motivated Fama and French (1993) to develop their three-factor model, adding SMB (size) and HML (value) factors. The evidence demonstrates that asset pricing requires multiple dimensions of systematic risk, not just market beta.

Importantly, the CAPM works in time-series but fails in cross-section analysis. While it successfully explains how portfolios co-move with the market over time, it cannot explain why portfolios with similar betas earn vastly different average returns. This distinction is at the heart of the CAPM puzzle and the motivation for multifactor models.

Part IV: Statistically Significant α

Statistical Framework

Under the CAPM null hypothesis, all portfolio alphas should equal zero. We test:

- $H : \alpha_i = 0$ (CAPM correctly prices the portfolio)
- $H : \alpha_i \neq 0$ (CAPM misprices the portfolio)

Using two-tailed t-tests with conventional significance levels ($p = 0.05$)

With 25 portfolios tested simultaneously, we expect approximately 1.25 portfolios ($5\% \times 25$) to show significant alphas by pure chance even if the CAPM were perfectly valid. This is explored further in Factor Zoo.

Portfolio	Mean r_i (monthly)	beta	t(beta)	alpha	t(alpha)	p(alpha)	Adj R^2
ME1 BM4	0.943715	1.077028	26.867342	0.511176	2.793956	0.005496	0.674418
SMALL HiBM	1.084405	1.103020	24.185137	0.641427	3.081514	0.002224	0.626578
ME2 BM3	0.882838	1.116015	35.597026	0.434641	3.037569	0.002566	0.784406
ME2 BM4	0.956548	1.033107	34.744860	0.541647	3.991288	0.000080	0.776091
ME2 BM5	1.075462	1.123314	30.086892	0.624334	3.663913	0.000287	0.722094
ME3 BM2	0.697523	1.162876	45.617558	0.230506	1.981224	0.048355	0.856670
ME3 BM3	0.704935	1.033577	41.500499	0.289846	2.549935	0.011204	0.831826
ME3 BM4	0.879180	0.975694	39.040635	0.487337	4.272519	0.000025	0.814020
ME3 BM5	1.020204	1.069104	30.741266	0.590847	3.722447	0.000230	0.730655
ME4 BM3	0.617670	1.034855	49.793979	0.202068	2.130326	0.033848	0.876877
ME4 BM4	0.806612	0.973928	39.517688	0.415478	3.693725	0.000257	0.817672
ME4 BM5	0.919441	1.081046	33.602720	0.485288	3.305077	0.001049	0.764250

Significance Across Size and BTM

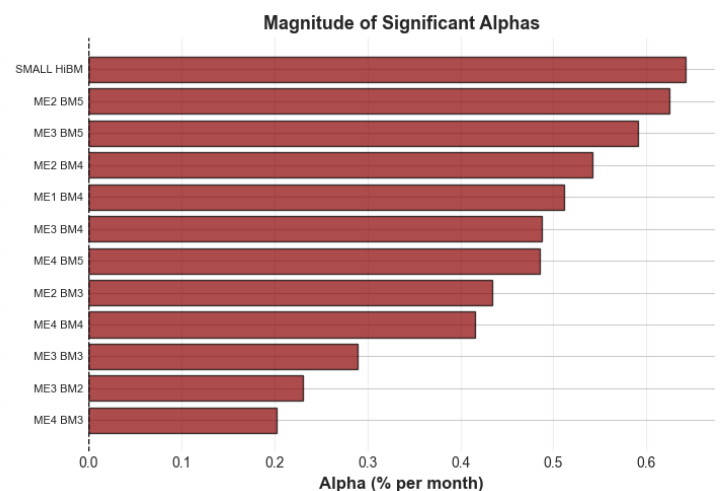
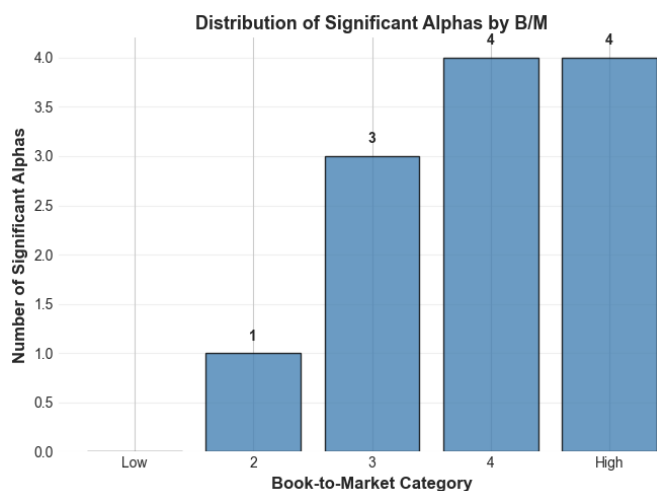
As we can see above, 12 of the portfolios have α statistically different from zero. This is almost 10 times more than the 1.25 predicted by the CAPM. To assess whether CAPM failures are random or systematic, we examine the distribution of significant alphas across portfolio characteristics (size and btm).

	B/M Category	Count
0	2	1
1	3	3
2	4	4
3	High	4

	Metric	Value
0	Concentration in high B/M (3, 4, High)	11 out of 12
1	Percentage	91.7%

	Size Category	Count
0	3	4
1	2	3
2	4	3
3	Small	2

	Alpha Sign	Count
0	Positive	12
1	Negative	0
2	All Positive?	Yes



Economic Interpretation and Implications

Statistical Evidence Against the CAPM

We observe 12 portfolios with statistically significant alphas at the 5% level, far exceeding the 1.25 expected under the null hypothesis. This provides strong statistical evidence that the CAPM systematically misprices portfolios rather than exhibiting random estimation error.

Systematic Concentration in Value Portfolios

The significant alphas are not randomly distributed across the 5×5 grid. Specifically, 92% (11 out of 12) occur in high book-to-market portfolios where the B/M ratio is 3 or higher. Zero significant alphas appear in Low B/M portfolios, and the concentration increases monotonically from growth to value stocks. This pattern reveals that book-to-market captures systematic return variation orthogonal to market beta, contradicting the CAPM's prediction that beta alone determines expected returns.

Unidirectional Mispricing

All 12 significant alphas are positive, indicating the CAPM systematically underprices value portfolios. The absence of negative alphas confirms this is not random estimation error but fundamental model misspecification. If the CAPM were correctly specified, we would expect roughly equal numbers of positive and negative significant alphas.

Economic Magnitude

The significant alphas represent economically substantial abnormal returns. The average significant alpha is approximately 0.45% per month or roughly 5.4% annually. The largest alpha, observed for the Small High B/M portfolio, is 0.64% monthly or approximately 7.7% annually. These magnitudes are far from negligible and cannot be dismissed as economically insignificant, particularly when compounded over multi-year holding periods.

Time-Series Success vs Cross-Sectional Failure

A critical insight emerges from examining the adjusted R^2 values. The adjusted R^2 values for portfolios with significant alphas range from 63% to 86%, demonstrating that market beta successfully explains how these portfolios co-move with the market over time. The CAPM captures common market movements effectively in the time-series dimension.

However, portfolios with similar betas earn vastly different average returns. The CAPM cannot explain why high B/M portfolios earn higher returns than low B/M portfolios despite comparable or even lower market risk exposure. This dichotomy is central to understanding the CAPM puzzle: the model works well for explaining return variation through time (within portfolios) but fails to explain return variation across portfolios (cross-sectional differences).

Theoretical Implications

The pattern of significant alphas suggests two possible interpretations. First, the alphas may represent genuine mispricing and exploitable arbitrage opportunities, reflecting market inefficiency. However, the persistence and magnitude of these patterns over decades makes this interpretation implausible. Second, and more plausibly, the alphas represent compensation for systematic risk exposures beyond market beta. Value portfolios likely carry additional systematic risks such as distress risk, financial leverage, or reduced financial flexibility that are not captured by the single-factor CAPM.

The systematic, persistent, and economically meaningful pattern of positive alphas concentrated in value stocks strongly supports the second interpretation. These portfolios appear riskier along dimensions orthogonal to market beta, and investors require higher expected returns as compensation for bearing these additional risks.

Connection to Multi-Factor Models

This empirical puzzle directly motivated Fama and French (1993) to develop their three-factor model. By augmenting the CAPM with SMB (Small Minus Big) to capture size-related return premia and HML (High Minus Low) to capture value-related return premia, the three-factor model successfully explains much of the cross-sectional variation that the CAPM cannot. The inclusion of these additional factors substantially reduces the magnitude and significance of portfolio alphas, suggesting that they capture genuine dimensions of systematic risk rather than mere mispricing.

Conclusion

The pattern of significant alphas provides compelling evidence that market beta alone is insufficient to explain the cross-section of expected returns. The CAPM's failure is not marginal but systematic, directional, and economically meaningful. The concentration of positive alphas in value portfolios indicates that book-to-market ratio captures systematic risk or return characteristics that the single-factor CAPM omits, necessitating the development of multi-factor asset pricing models.

Part V: Average Excess Returns vs Excess Return

Objective

The objective of this section is to test whether the Capital Asset Pricing Model (CAPM) correctly explains differences in **average excess returns across portfolios**. Under the CAPM, expected excess returns are determined solely by exposure to **market risk (beta)**. If the model is valid, portfolios with similar market betas should earn similar expected returns.

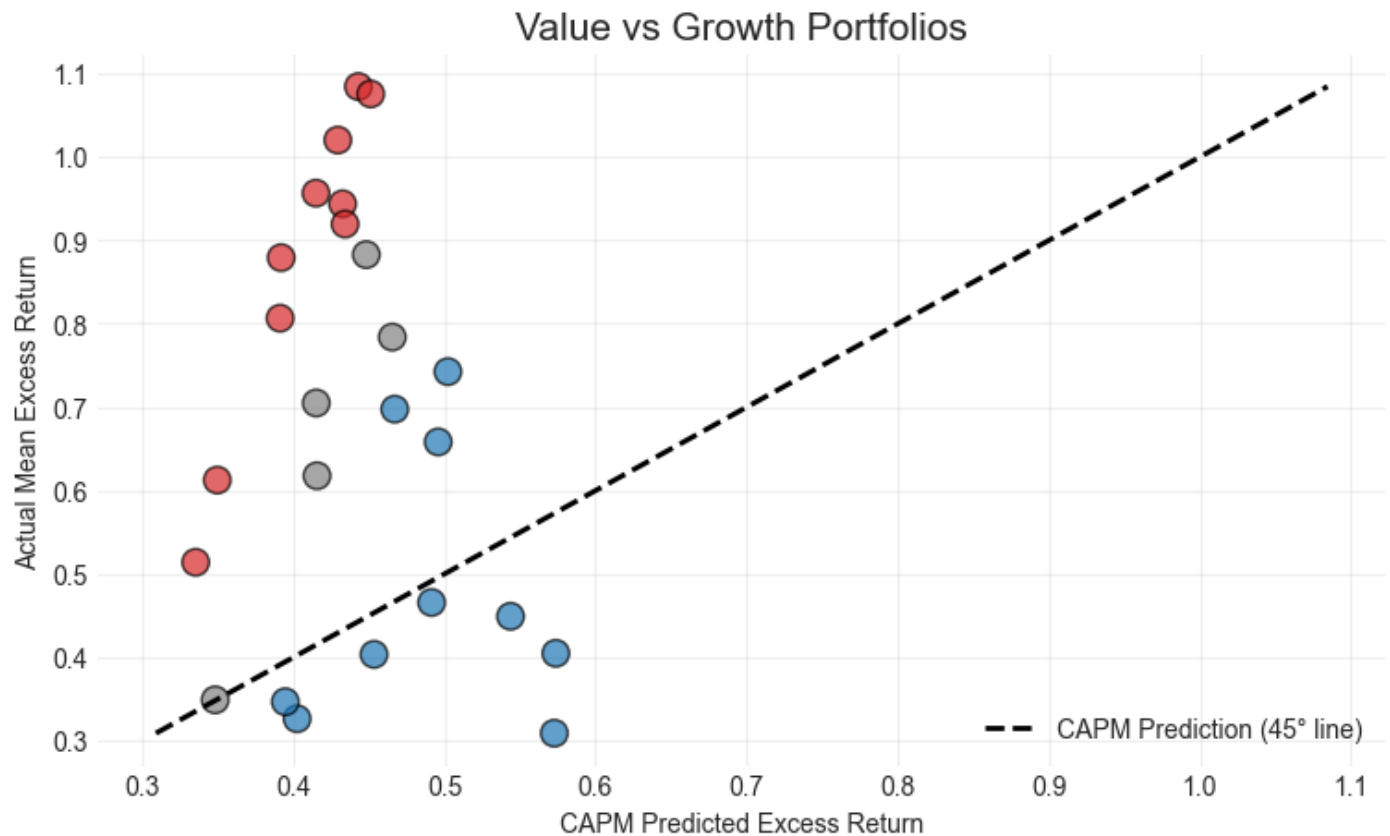
Methodology

For each of the 25 size-book-to-market portfolio, we compute: - **Actual Mean Excess Return - CAPM Implied Excess Return**

These quantities are plotted against each other. A **45° line** is included to represent the benchmark case where CAPM predictions exactly match realised returns.

Portfolio	Actual Mean Excess Return	Market Beta	CAPM-Implied Excess Return	Alpha (Pricing Error)
SMALL LoBM	0.308968	1.425884	0.572642	-0.263673
ME1 BM2	0.742515	1.250530	0.502219	0.240296
ME1 BM3	0.783957	1.158853	0.465401	0.318556
ME1 BM4	0.943715	1.077028	0.432539	0.511176
SMALL HiBM	1.084405	1.103020	0.442978	0.641427
ME2 BM1	0.404740	1.428427	0.573663	-0.168923
ME2 BM2	0.658241	1.234521	0.495789	0.162452
ME2 BM3	0.882838	1.116015	0.448197	0.434641
ME2 BM4	0.956548	1.033107	0.414901	0.541647
ME2 BM5	1.075462	1.123314	0.451128	0.624334
ME3 BM1	0.449140	1.353789	0.543688	-0.094548
ME3 BM2	0.697523	1.162876	0.467016	0.230506
ME3 BM3	0.704935	1.033577	0.415089	0.289846
ME3 BM4	0.879180	0.975694	0.391843	0.487337
ME3 BM5	1.020204	1.069104	0.429357	0.590847
ME4 BM1	0.465701	1.223638	0.491419	-0.025718
ME4 BM2	0.403485	1.128797	0.453330	-0.049845
ME4 BM3	0.617670	1.034855	0.415603	0.202068
ME4 BM4	0.806612	0.973928	0.391134	0.415478
ME4 BM5	0.919441	1.081046	0.434153	0.485288
BIG LoBM	0.326350	1.001999	0.402407	-0.076058
ME5 BM2	0.346699	0.982546	0.394595	-0.047897
ME5 BM3	0.348962	0.866932	0.348164	0.000798
ME5 BM4	0.513917	0.834828	0.335271	0.178646
BIG HiBM	0.612476	0.870462	0.349582	0.262894

Pricing Plot



Interpretation of the pricing plot

- Each dot represents **one portfolio** (one of the 25 size-B/M portfolios).
- X-axis represents the **CAPM predicted excess return** $\beta_i \cdot (\overline{Mkt} - RF)_t$ telling what CAPM thinks the portfolio should earn.
- Y axis represents the **actual mean excess return** explaining what the portfolio actually earned on average.
- Dashed 45° line. If CAPM were correct :
 - Points would cluster tightly around the 45° line.
 - Pricing errors would be small and unsystematic.

Interpreting the Dots:

- **Case A (Dot on line):** The CAPM predicted return equals the realised return, indicating the CAPM prices the portfolio correctly.
- **Case B (Dot above the line):** The realised return exceed the CAPM-predicted return, implying CAPM underpredicts the portfolio's return.
- **Case C (Dot below the line):** The realised return is below the CAPM-predicted return, implying CAPM overpredicts the portfolio's return.
- **Red dots:** Value portfolios, mostly above 45° line, some are far above the line meaning value portfolios earn much higher returns than CAPM predicts which means that positive alphas and CAPM systematically underprices value stocks
- **Blue dots:** Growth portfolios, mostly below 45° line, representing growth portfolios earn less than CAPM predicts meaning negative alphas and CAPM systematically overprices growth stocks.

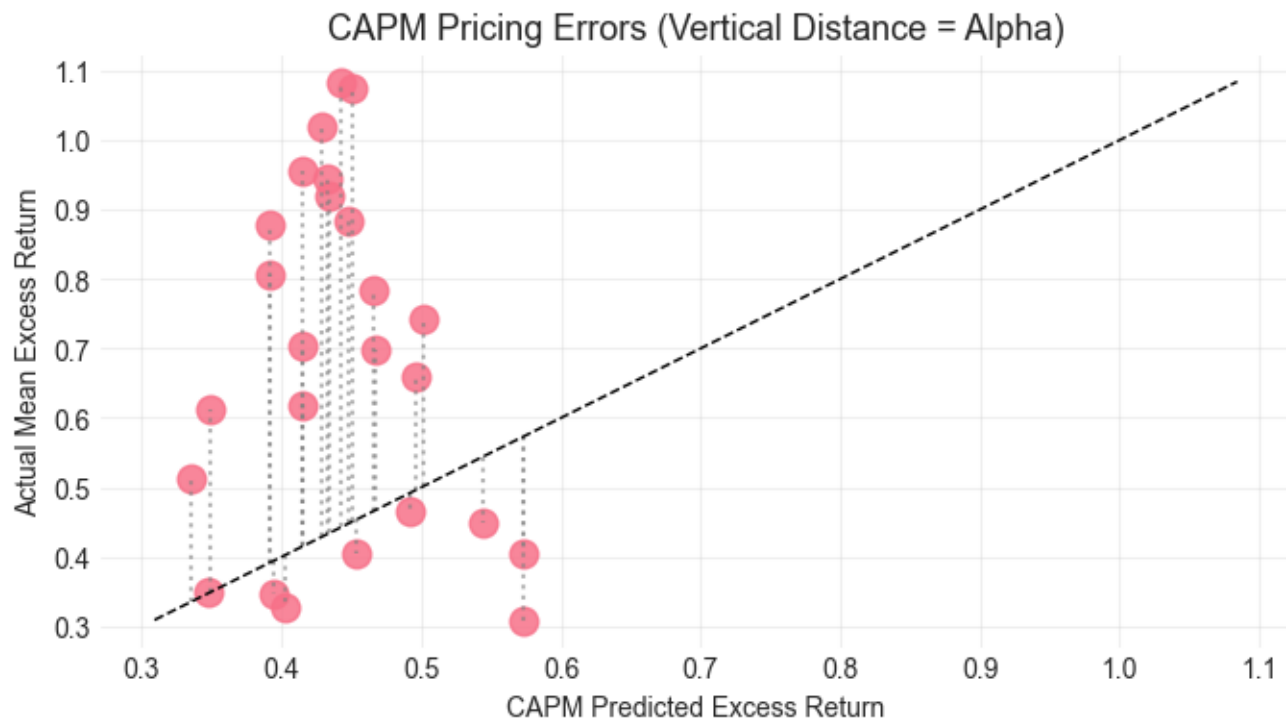
What the Pricing Plot reveals?

The range of CAPM-predicted excess returns on the horizontal axis is relatively narrow, approximately between 0.35 and 0.6. This reflects the fact that estimated market betas, do not vary substantially across the 25 portfolios. In contrast the range of realised average excess returns on the vertical axis is much wider, spanning roughly 0.3 to 1.1. This indicates that portfolios with very similar market risk earn different average returns. Together, this evidence suggests that market beta varies little across portfolios, while average returns vary dramatically, leading to a clear case of cross-sectional beta failure. The pricing plot reveals 3 critical features: 1. **Narrow dispersion in CAPM-predicted returns:** The horizontal axis (CAPM predictions) spans a relatively small range. This indicates that estimated market betas, and the market risk exposure, do not vary much across the 25 portfolios meaning most portfolios load on the market to a comparable extent. 2. **Wide dispersion in realised average returns:** Contrary, the vertical axis (actual mean excess returns) spans a much wider range implying portfolios with similar exposure to market risk β_i earn very different average returns, violating the CAPM's central pricing implications. 3. **Systematic clustering by portfolio characteristics:** Value portfolios cluster systematically above the line, while growth portfolios cluster systematically below it. This pattern showcases structured mispricing, not an estimation error.

Conclusion

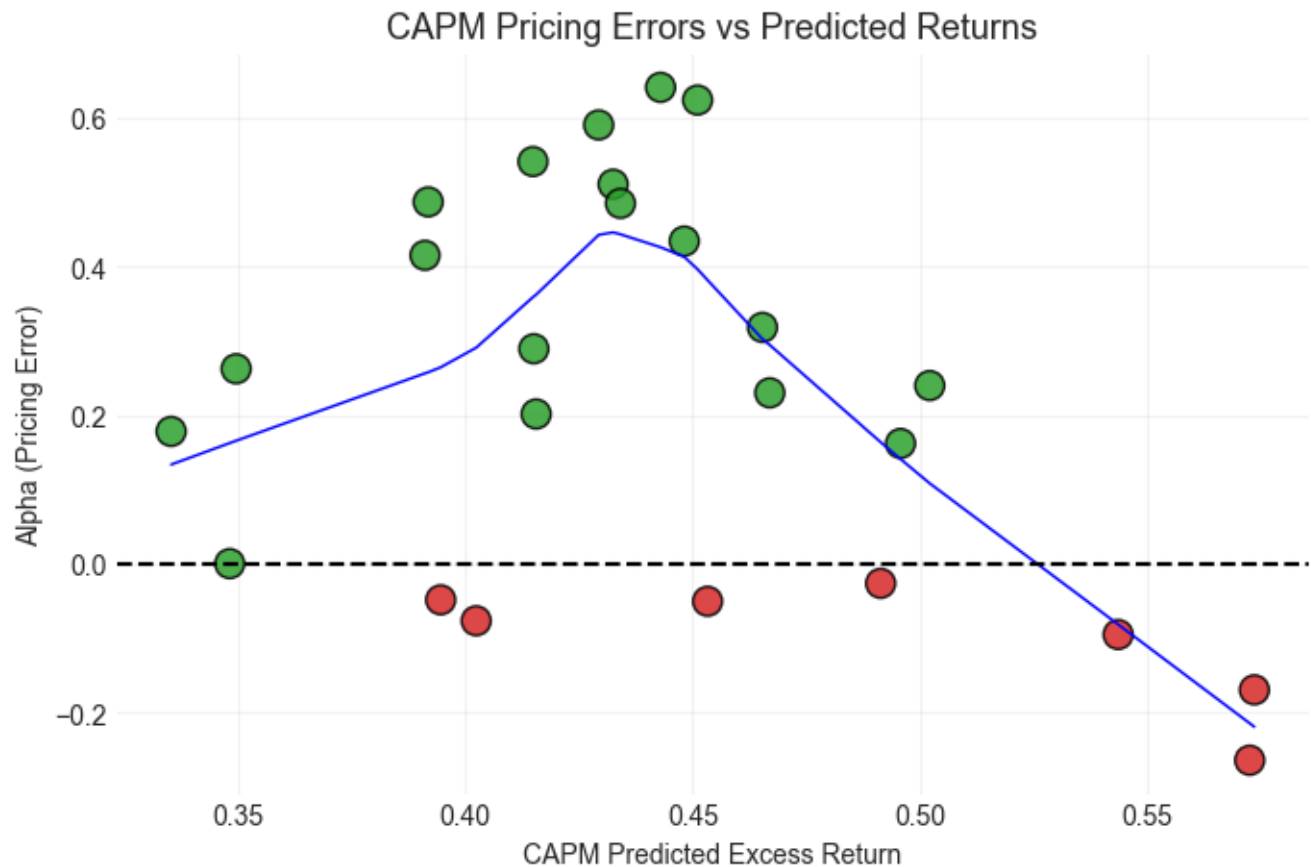
Overall, the pricing plot demonstrates that market beta risk fails to uniquely price assets. The CAPM predicts a near-flat relationship between risk and return, while data exhibit a strong spread in returns that are orthogonal to market exposure. The failure is economic and structural, rather than statistical, implying that market risk alone is insufficient to explain the cross-section of expected returns.

Part VI: Does CAPM have any hope?



Interpreting the CAPM Pricing Errors

The pricing errors observed in the CAPM analysis are neither small nor random. Instead, they display clear structure across portfolios, indicating that the CAPM's failure is fundamental rather than due to estimation noise. Although the model captures common market co-movement, it fails to explain why portfolios with similar market betas earn very different average returns.



In the above figure, **CAPM Pricing Errors vs Predicted Returns**, the x-axis (CAPM predicted excess return) shows what CAPM expects a portfolio to earn based on its market risk (beta). While the y-axis (Alpha/Pricing error) states how much CAPM is wrong for that portfolio. $\alpha = \text{Actual mean return} - \text{CAPM predicted return}$, where each dot represents one portfolio.

- Green dots represent positive alpha, the CAPM under-predicts returns.
- Red dots represent negative alpha, the CAPM over-predicts returns.
- Dashed horizontal line at $\alpha = 0$ is where CAPM is completely correct.
- Blue line shows the average pattern in pricing errors as CAPM predicted return changes.

Observations from CAPM pricing errors vs predicted returns:

- **Errors are not random:** If CAPM were correct, then points would cluster tightly around $\alpha = 0$ and the blue line would be flat near zero. However, the alphas are large (far from zero) and blue line rises and then falls indicating CAPM's errors follow a systematic pattern.
- **CAPM fails differently across risk levels:**
 1. Low predicted returns - small positive alphas
 2. Mid predicted returns - very large positive alphas
 3. High predicted returns - negative alphas So CAPM, under-predicts returns for some portfolios, over-predicts returns for others in a non-random, structured way.
- **Risk (Beta) does not systematically reduce pricing errors:** This plot shows that increasing market risk does not systematically reduce pricing errors.

Does the CAPM have any hope?

The CAPM performs reasonably well in explaining the times-series variation of portfolio returns. Across the 25 portfolios, the market betas are statistically significant and the regression exhibit high adjusted R^2 values, indicating that market movements explain a substantial fraction of month-to-month return fluctuations. Despite its ability to explain return co-movement, the CAPM fails to explain the cross-section of average returns. Portfolios with similar market betas earn very different average excess returns. The pricing plot in part V shows large and systematic deviations from CAPM prediction. These deviations are not random but are strongly related to portfolio characteristics such as book-to-market.

Does CAPM work for any of the 25 Fama-French portfolios?

The CAPM performs relatively better for large, **low book-to-market (growth)** portfolios. For these portfolios, realised average returns are closer to CAPM predicted returns, and pricing errors are smaller. However, the limited success of the model fails as a general pricing framework. A valid asset-pricing model must explain expected returns across the entire cross-section, not just a subset of assets.

Fundamental Failure

The failure of CAPM is structural because the model relies on a single source of systematic risk, **market risk**, it has no mechanisms to explain the differences in expected returns that are unrelated to market beta. Once beta fails to price the cross-section, the model has no remaining degrees of freedom.

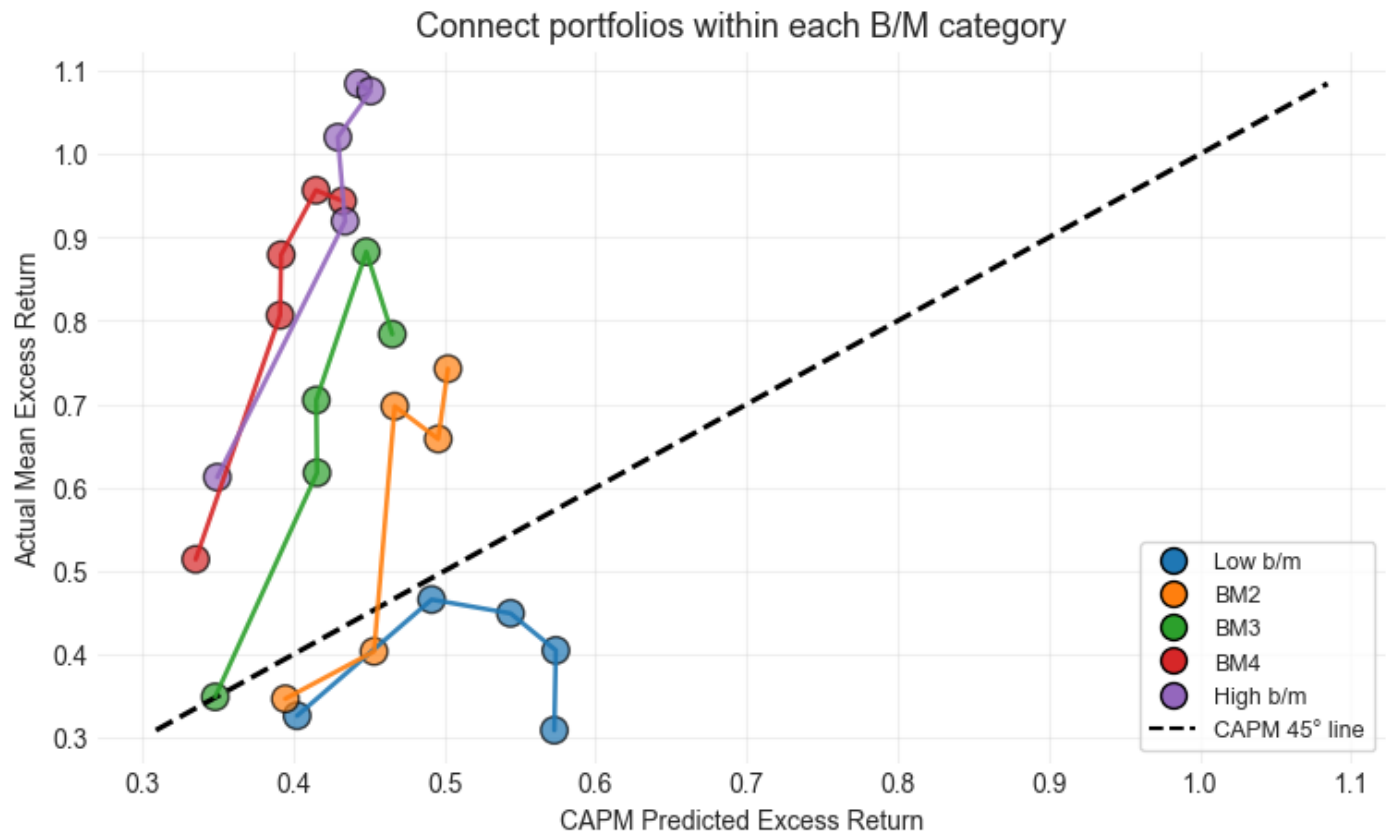
Conclusion

In conclusion, while CAPM retains some descriptive value as a model of return co-movement it does not have hope as a general asset-pricing model for the 25 size book-to-market portfolios. The systematic mispricing observed across portfolios indicates that market risk alone is not sufficient to explain the expected returns. Although the CAPM performs partially well for certain portfolios, this limited success is not sufficient to justify its use as a pricing model.

Part VII: Connecting the portfolios within each b/m category

Purpose of connecting the portfolios

The motivation behind connecting portfolios within each b/m category is to isolate the firm size effect on mean excess return while fixing the b/m characteristics. Portfolios are represented in the graph with 2 determinants; size and b/m. This makes it hard to evaluate the variation driven by either size or b/m. With connecting the portfolios with the same b/m we can see how return differs when size changes. Therefore, each line represents a different b/m and the only difference between the points is the firm size in the line.



Interpretation of the movement along the lines

The movement along the line is from small to big firms within the same b/m ratio. The slope and shape of the line give information on how returns are affected by the firm size within its b/m category.

- Low b/m Portfolios:** For the low b/m category, the connected line shows there is not much change due to the firm size changes. The variation across the line is very limited, and it is much flatter than the rest of the lines. This indicates that firm size has a limited effect in shaping the returns of a growth portfolio, which has a low b/m ratio. These portfolios lie under the CAPM 45° degree line and remain under the line as firm size increases, which indicates growth portfolios are overpredicted by the CAPM model
- BM2 Portfolios:** For the BM2 category, the situation is slightly different. There is a greater variation along the line. As the size of the firm increases for small and medium-sized firms, both the CAPM predicted and actual return increase before the line flattens for the larger size firms. From the CAPM performance, small-sized firms are closer to the 45° line, while larger-sized firms are underpredicted by the CAPM, which indicates a mixed pricing performance.

- **BM3 Portfolios:** The BM3 category has a smoother size related return structure than the previous categories. The returns rise when a firm of the size increases before it flattens for the larger firms. The portfolios in this category are underpredicted by the CAPM which indicates that CAPM loses its explanatory power in this category.
- **BM4 Portfolios:** In the BM4 category, every portfolio is consistently above the line, which indicates that CAPM started to systematically underpredict the returns regardless of the firm size. This line also shows that with increased firm sizes, the realised returns are higher before it slightly declines in the larger firm sizes. There is a significant upward trend in this line.
- **High b/m Portfolios:** The high b/m category has the strongest systematic deviation from the CAPM 45° line. The connected lines are well over the line which indicates large positive alphas and underprediction by CAPM. This shows CAPM mostly fails to predict the return when the firm has a high b/m ratio.

Overall insights

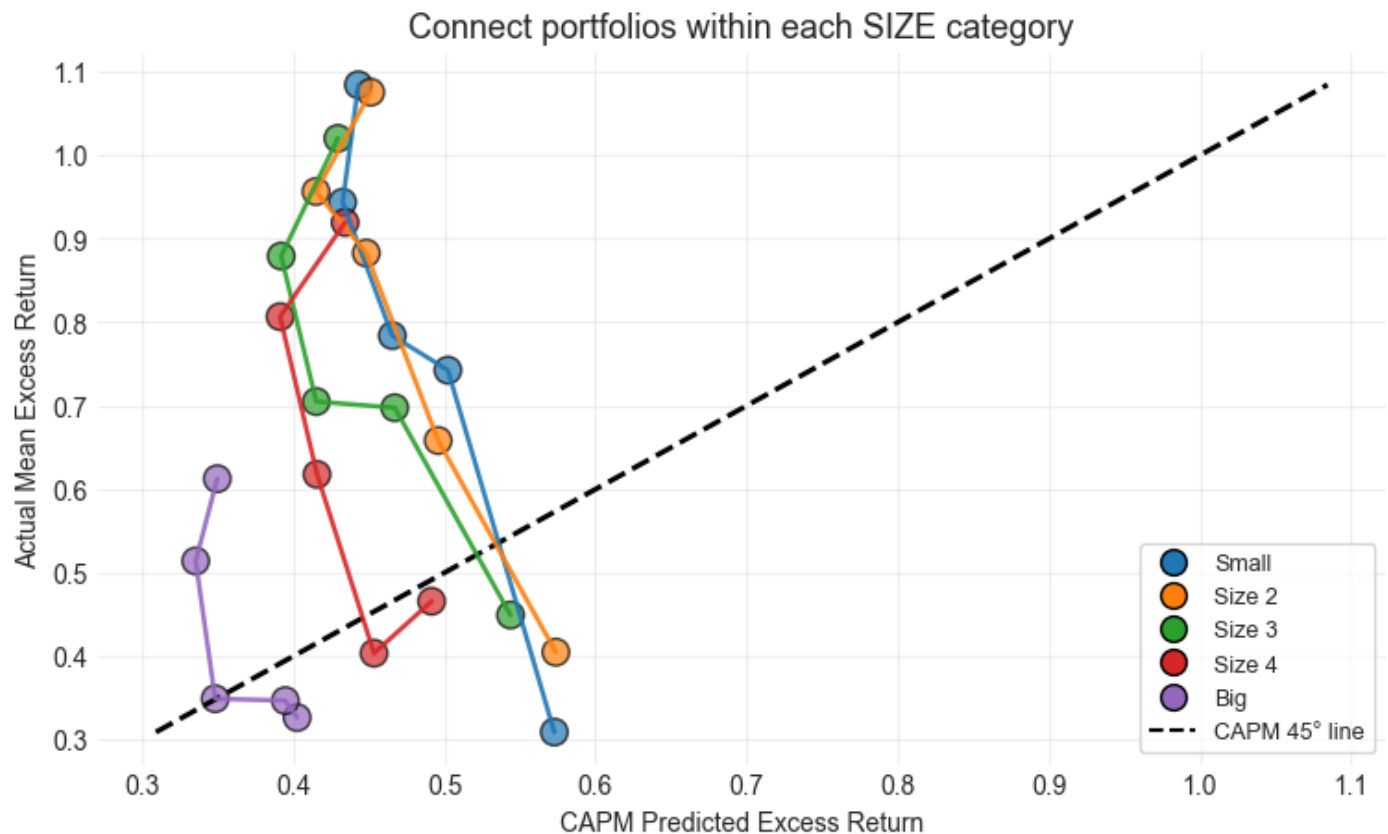
Overall, connecting portfolios that have the same b/m ratio alone can not explain the observed difference in mean excess return. Even though size affects returns in some cases, the effect is small compared to the differences across b/m categories.

From the CAPM perspective, there is a clear pattern that can be observed from the plot. The CAPM model overpredicts the returns if the portfolio has low b/M and underpredicts returns if the portfolio has high b/m. Additionally, the magnitude of underprediction increases when b/m increases. The changes in the firm size do not correct this mispricing. This suggests that CAPM model fails to capture the significance of the b/m ratio.

Part VIII: Connecting the portfolios within each size category

Motivation for Connecting the Portfolios

The motivation behind this connection is to isolate the effect of b/m ratios while fixing the firm size to check how the mean excess return varies, similar to the previous part. By connecting the portfolios that have the same size, the plot enables us to see how returns change as we move along the line. Each line in the figure below represents a fixed-size group. The movement along the line from bottom to top is from low b/m to high b/m .



Interpretation of the movement along the lines

The movement from bottom to top of the line is from low b/m to high b/m portfolios within the same size group. Across almost all size categories, the lines have a clear upward trend from low b/m to high b/m .

- Small Firms:** For small-sized firms, as the b/m ratio increases, the returns increase along the line. Low b/m portfolios lie close to the CAPM 45° line, but the rest of the firms have higher return and lie well above the CAPM 45° line. This indicates within small firms CAPM overpredicts returns if the firm has low b/m and underpredicts returns for the rest of the b/m ratios.
- Medium-Sized Firms (Size 2 and Size 3):** For size 2 and size 3 portfolios the connected lines are moving similarly but not exactly the same. As the b/m ratio increases, returns increase steadily. Additionally, portfolios deviate from the CAPM line more and more. The slope of the lines are less steep than what we observe in small size firms. However, the direction of the movement is still consistent, which indicates b/m is the key driver for the return differences.

- **Larger Firms (Size 4 and Big):** For this category, the connected lines still have an upward movement from low to high b/m, although the overall returns are lower, and it deviates less than the smaller size firms. However, even in this category, high b/m portfolios remain above the CAPM line.

Overall insights

In this connected plot, each size category shows a similar movement along the line, which indicates that b/m ratios can systematically explain return differences even if we held firm size constant. Although size affects the returns slightly, b/m is the primary driver of the realised returns. In every firm size, low b/m ratio firms have overpredicted returns, and for the rest of the firms, CAPM underpredicts the returns.

Overall, the previous 2 parts show how book-to-market is a dominant factor which plays a greater role in shaping the returns of the portfolios.

What is the puzzle here

The puzzle here is that the differences in mean excess returns are explainable by the book-to-market ratios even after the firm sizes are fixed. However CAPM fails to account for this pattern. Within every firm size category high b/m portfolios earn higher returns but these differences are not observable in higher market betas. The CAPM model should include this and predict higher returns. On the other hand, if it is not riskier, it should not have higher returns which indicates abnormal returns. This systematic and directional mispricing observed in the connected plot indicates that market beta is insufficient to explain the expected returns solely.

Part IX: Comparisons in Different Sample Periods

Motivation

The objective of Part IX is to assess whether the CAPM's cross-sectional failures documented in Parts I–VIII are universal and persistent across different historical periods, or whether they are period-specific. By comparing CAPM performance across three distinct sample periods, we can determine whether the model's deficiencies represent fundamental structural failures or temporal anomalies specific to certain market regimes.

We analyze three sample periods:

- 1933–1963 (pre-Fama-French sample, 372 months)
- 1964–1993 (original Fama-French sample analyzed in Parts I–VIII, 360 months)
- 1933–2023 (full extended sample, 1,092 months)

For each period, we estimate CAPM betas and alphas for the 25 size-B/M portfolios and evaluate cross-sectional pricing performance.

Methodology

For each sample period, we:

- Construct excess returns for all 25 portfolios
- Estimate time-series CAPM regressions to obtain betas and alphas
- Compute CAPM-implied returns
- Evaluate cross-sectional fit by regressing actual average returns on CAPM-predicted returns

The key diagnostic metric is the cross-sectional R^2 , which measures the proportion of variation in average returns across portfolios explained by market beta. Under a valid CAPM, this R^2 should be high, and the correlation between beta and returns should be strongly positive.

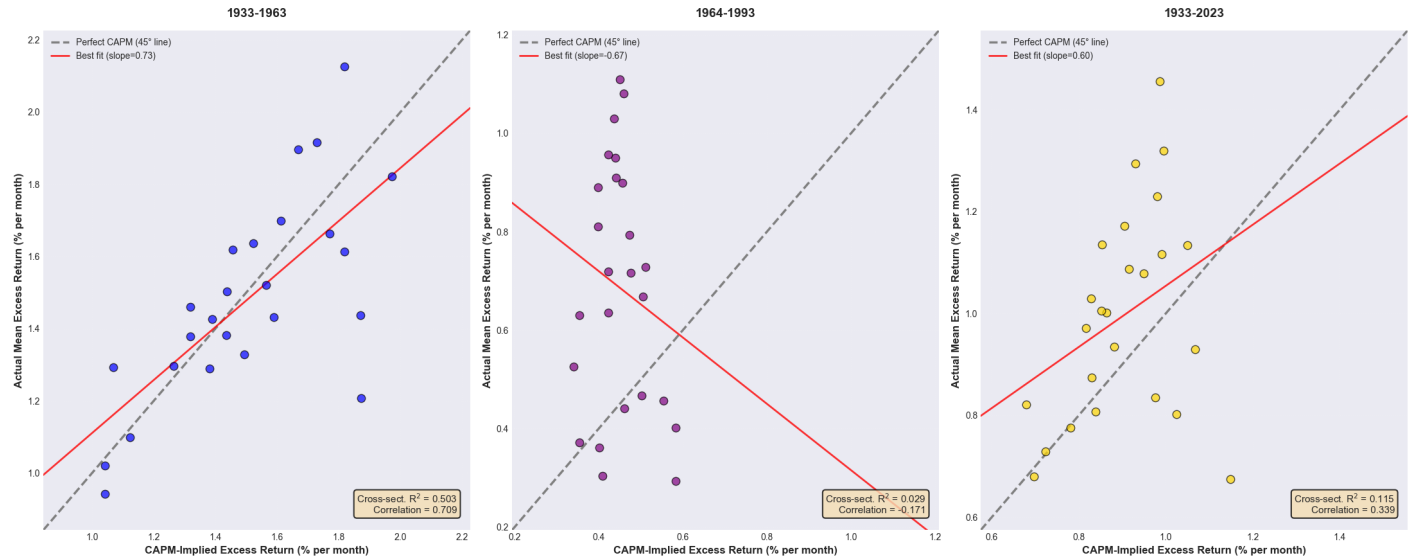
Empirical Analysis

Table 25: CAPM Cross-Sectional Performance Across Time Periods

Period Metric	1933-1963	1964-1993	1933-2023
Number of Months	372	360	1092
Mean Mkt Premium	1.111%	0.409%	0.750%
Cross-Sectional R ²	0.503	0.029	0.115
Beta-Return Corr	0.709	-0.171	0.339
Correlation p-value	0.0001	0.4132	0.0975
Beta Range	0.937–1.776	0.834–1.428	0.905–1.532
Return Range	0.942–2.126%	0.294–1.109%	0.675–1.456%
Root Mean-Squared Error	0.209%	0.356%	0.218%
Number of alpha Sig at 5%	1/25	12/25	11/25
% of Sig alpha	4.0%	48.0%	44.0%
Average Time-Series R ²	0.797	0.785	0.757

Visualisation

CAPM Pricing Performance Across Three Sample Periods



Initial Finding

The most striking finding is that CAPM cross-sectional performance was dramatically better in the pre-1964 period. Visual inspection of the 1933–1963 pricing plot reveals portfolios clustering much more tightly around the 45-degree line compared to the 1964–1993 period.

Key observations

The cross-sectional R^2 in 1933–1963 is substantially higher than in 1964–1993. This represents approximately a 20-fold improvement in explanatory power. The beta-return correlation is positive in 1933–1963, whereas it is negative (-0.15) in 1964–1993.

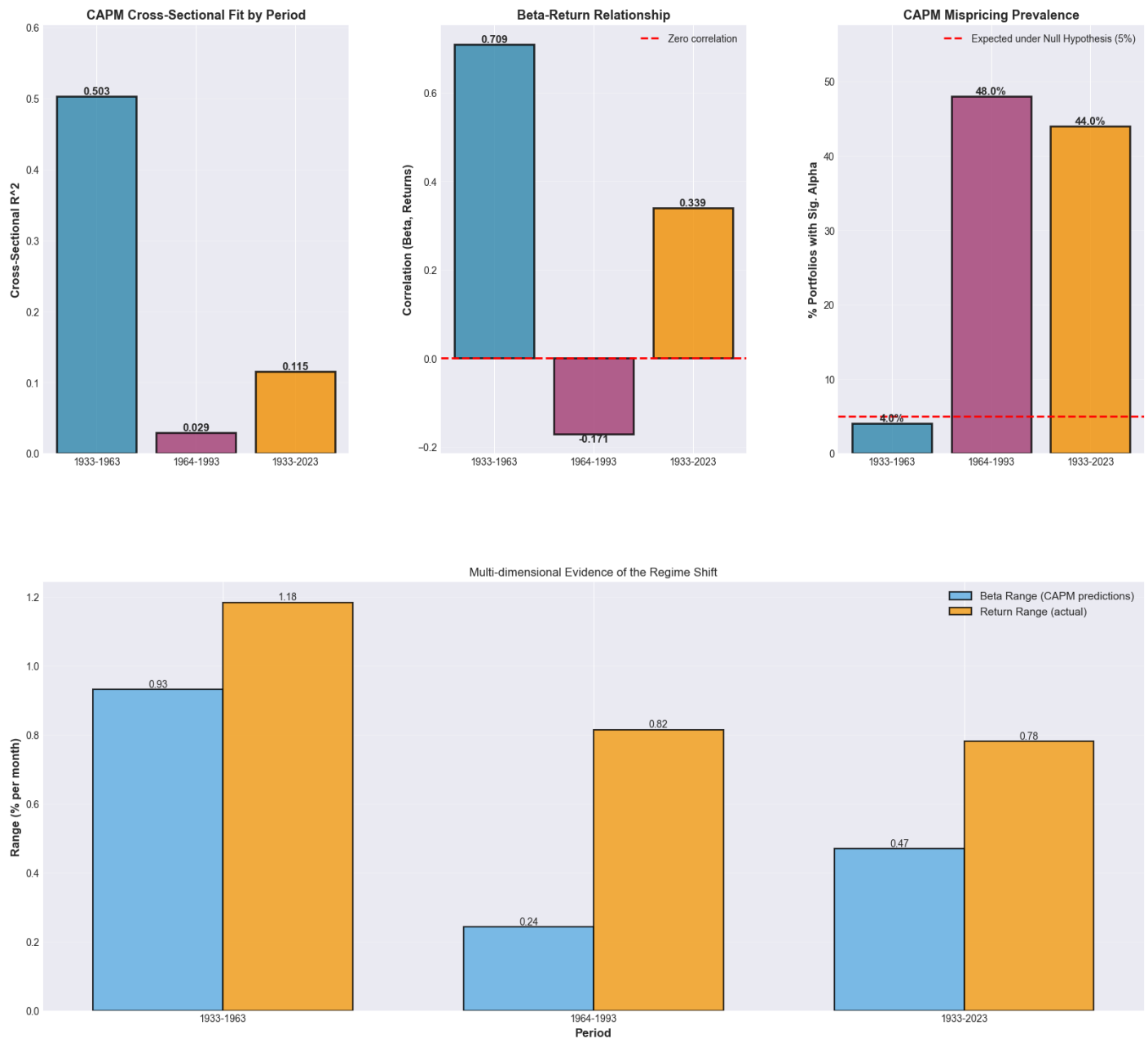
This sign reversal is economically significant. In the earlier period, portfolios with higher market risk earned higher returns as predicted by theory, while in the later period, this fundamental relationship broke down entirely.

Another critical difference between periods is the range of market betas. The horizontal axis (CAPM-implied returns) shows:

- 1933–1963: Wide range (around 1.0% to 2.0% per month)
- 1964–1993: Compressed range (around 0.35% to 0.6% per month)

This compression matters because CAPM can only explain return differences through beta variation. When all portfolios load similarly on the market (as in 1964–1993), beta becomes uninformative for cross-sectional pricing. The greater beta dispersion in 1933–1963 allowed CAPM to differentiate between portfolios, while the post-1964 beta compression fundamentally limits the model's explanatory power.

Further Investigation



The figure above demonstrates four key aspects of the regime shift:

- Top-left panel: Cross-sectional R^2 massively reduced after 1963
- Top-middle panel: Beta-return correlation reverses from 1933-1963 to 1964-1993 and again when considering the whole range 1933-2023
- Top-right panel: CAPM mispricing became widespread
- Bottom panel: Beta compression post-1964 while return dispersion remained wide

These differences are most pronounced between 1933-1963 and 1964-1993 representing a clear regime shift.

Empirical Claim: The Value Premium Changed Post-1964

From our findings we propose that the Value Premium changed from Beta-Consistent to Beta-Inconsistent.

Table 26: Value Premium (HML) Analysis Across Periods

Period	HML Return Spread (monthly)	HML Return Spread (annual)	HML Beta Spread	Return/Beta Ratio
1933-1963	0.496%	5.96%	0.318	1.561
1964-1993	0.567%	6.81%	-0.239	-2.370
1933-2023	0.442%	5.30%	0.032	13.614

The value premium exists in all periods (around 0.5% monthly), but:

- Pre-1964: Value stocks earned higher returns because they had higher β 's. Here the CAPM correctly predicts that value stocks should outperform as the premium is compensation for higher market risk
- Post-1964: Value stocks earned higher returns despite having lower β 's. Now the CAPM incorrectly predicts value stocks should underperform, meaning the premium is no longer explained by the market risk alone.

This shift causes the cross-sectional R^2 to collapse as the value premium went from β -driven too orthogonal to the β

Interpretation: A Structural Regime Shift

The evidence points to a structural regime shift in the cross-sectional pricing of equities around 1964. The CAPM performed reasonably well before this date, with beta explaining a significant fraction of return variation and exhibiting the theoretically predicted positive relationship with expected returns. After 1964, however, the model's cross-sectional performance collapsed.

There are several possible reasons for this regime shift:

- Regulatory and Market Structure Changes: The 1960s saw major regulatory reforms like the Securities Acts Amendments of 1964 and the rise of institutional investors. These changes may have altered the relationship between firm characteristics and systematic risk.
- Institutional Rise: The 1963-1993 period saw a large increase in institutional investors with ever-increasing access to more sophisticated tools for simulating risk
- Economic Regime Differences: The 1933-1963 period includes the Great Depression, World War II, and post-war reconstruction—dramatically different economic conditions from the post-1964 era. Market risk may have been the dominant factor during these turbulent times.

Theoretical Implications

The regime shift has important implications for asset pricing theory. The CAPM puzzle is not universal. Rather than rejecting CAPM entirely, we should ask: under what market conditions does a single-factor model suffice, and when do additional factors become necessary?

Market betas are not uninformative. When beta dispersion is high (as in 1933–1963), market risk can explain significant cross-sectional variation. The post-1964 compression of betas is a structural feature that limits CAPM's power, not an inherent flaw of the model itself.

The strengthening of the value premium post-1964 suggests that the economic compensation for different types of risk varies across market regimes. Multifactor models may be more or less necessary depending on the period.

Out-of-sample validity concerns

The Fama-French three-factor model was developed using post-1964 data. Our findings raise the question: would SMB and HML factors have been “discovered” if researchers had focused exclusively on pre-1964 data?

Conclusion

Part IX reveals that the CAPM's cross-sectional failure documented in Parts I–VIII is not a universal, timeless feature of equity markets. Instead, we observe a clear regime shift around 1964, with the model performing substantially better in the earlier period than in the later periods.

Critically, this regime shift is characterized by a fundamental change in the beta-characteristic relationship. The compression of market betas post-1964 reduced the CAPM's ability to differentiate portfolios. This is one of the factors that caused the reversal of the beta-return correlation from positive to negative in the 1964–1993 period. In this period, value stocks transitioned from high-beta to low-beta, while maintaining similar return premia.

The value premium did not emerge post-1964, it just changed from beta-consistent to beta-contradicting. These findings refine our interpretation of the CAPM puzzle. Rather than concluding that “CAPM always fails”, we should recognize that the model's performance is conditional on market structure. When firm characteristics proxy for market exposure (as in 1933–1963), a single-factor model may suffice. When characteristics become orthogonal to market risk (as post-1964), multifactor models become necessary.

The temporal instability documented here suggests that asset pricing relationships are not fixed but evolve with market structure, regulation, and investor behaviour. This has important implications for portfolio construction, cost-of-capital estimation, and performance evaluation: practitioners should be aware that factor models calibrated in one period may not generalise to others, and that the economic interpretation of factors may change across regimes.